# Infosys - Coding 9

HackerRank

You are given an array A of length N.

The value of a subsegment is calculated by multiplying all the values in that subsegment, for example the value of [2, 3, 5] is 2 * 3 * 5 = 30.

Find the total number of ways to partition the array A into consecutive subsegments that cover all elements such that the value of each subsegment is divisible by at least X factors. Since the number of ways can be large, return it modulo 10^9+7.

**Input Format**

The first line contains an integer, N, denoting the number of elements in A.

The next line contains an integer, X, denoting the number of factors the value of a subsegment must have.

Each line i of the N subsequent lines (where 0 ≤ i < N) contains an integer describing A[i].

**Constraints**

2 <= N <= 10^5

2 <= X <= 10^5

1 <= A[i] <= 10^5

**Output Format**

The output contains an integer value

**Sample Input 0**

```
2
2
2
2
```

**Sample Output 0**

```
2
```

**Explanation 0**

N=2, X=2, A=[2,2] There are two ways to partition the array: • [2], [2]. Product of each = 2, which is the value. The factors of 2 are 1, 2. • [2, 2]. 2*2 = 4. The number of factors of 4 are more than 2.

**Sample Input 1**

```
3
5
2
6
8
```

**Sample Output 1**

```
1
```

**Explanation 1**

N=3, X=5, A=[2,6,8] The only way is to contain the whole array as a subsegment. 268 = 96. Number of factors of 96 are more than 5.